

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q21-1664

Noti Creek 100 kV
October 03, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1664. The Interconnection Customer proposes a 174.9 MW wind facility interconnecting at the Noti Creek 100 kV substation in the Illinois area. The request was received on February 07, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 461 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$29,191,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Gvrd Sewage to Noti Creek 100 KV Line
Contingency: Noi Crk to W Amesbury 100 KV
Loading: 115.5% (Rating: 128 MVA)
DFAX: 100.0%
Pre-project loading: 21.5%
2. Facility: W Amesbury to Noti Crk 100 KV
Contingency: Gvrd Sewage to Ann 100 KV
Loading: 102.0% (Rating: 151 MVA)
DFAX: 100.0%
Pre-project loading: 14.1%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Gvrd Sewage to Noti Creek 100 KV Line	Rebuild line with higher-capacity conductor	\$16,255,000	100% (\$16,255,000)
W Amesbury to Noti Crk Substation 100 KV	Reconductor line	\$12,936,000	100% (\$12,936,000)
	Total allocated to project		\$29,191,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.